Smart Campus Management System

CS-104L Object-Oriented Programming

HITEC University Taxila

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Introduction

This project is a simple command-line Smart Campus Management System written in C++. It manages basic campus records such as students, faculty, courses, library items, fee records, hostel rooms, and reports. The code is made with beginner-level OOP concepts and simple arrays so it can be understood and explained by second-semester students.

System Modules

- Person module: Person, Student, GradStudent, Faculty, and Staff classes.
- Course module: Course and Enrollment classes with capacity checking.
- Library module: Book and Journal catalog using file handling and arrays.
- Finance module: FeeRecord and Invoice classes with copy handling and static invoice counter.
- Hostel module: Room, HostelBlock, and HostelManager using composition and multiple inheritance.
- Reports module: student sorting and campus text report generation.

Class Diagram

The class diagram below shows the main classes and relationships used in the project.

Smart Campus Management System - Class Diagram

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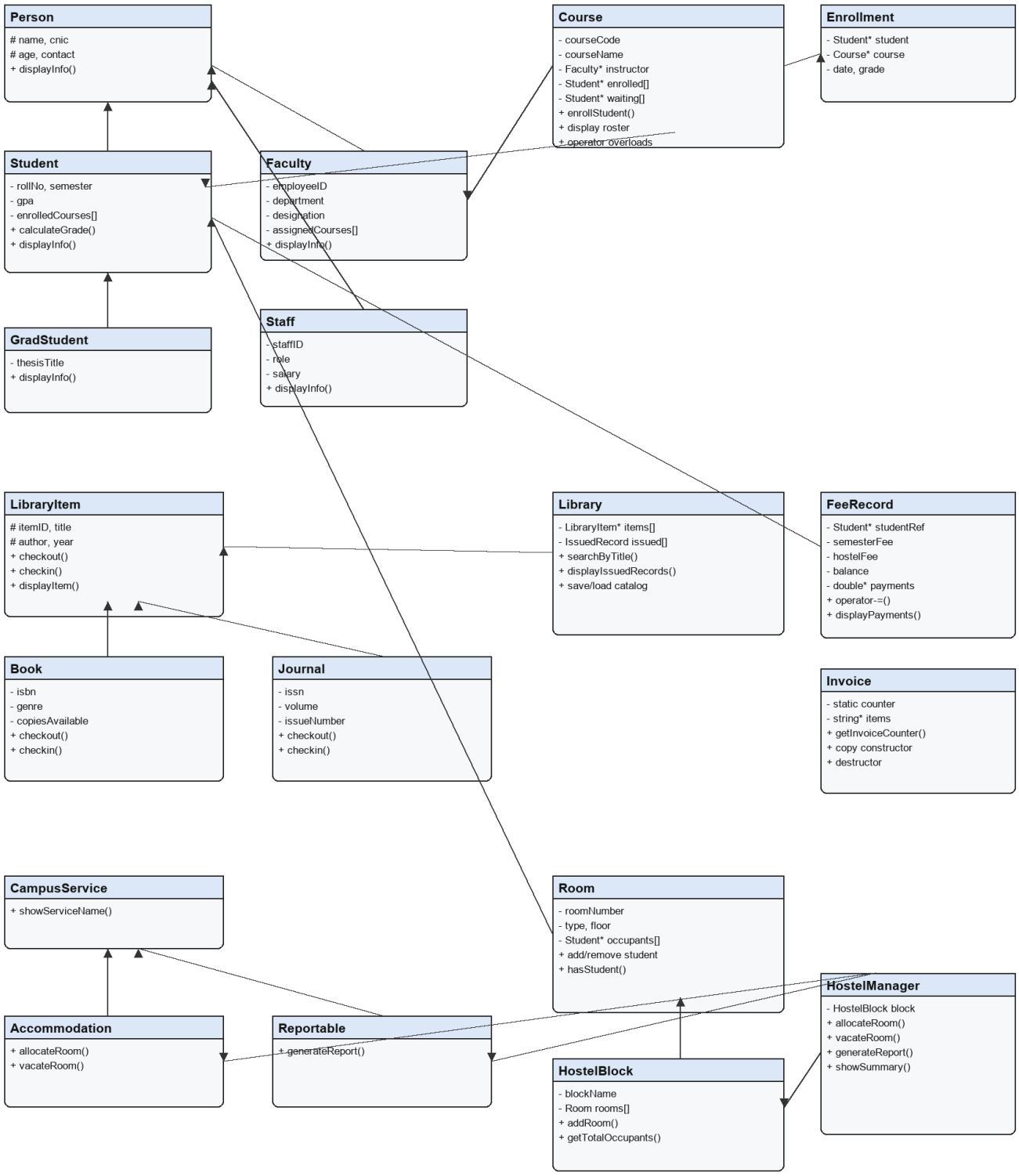

Legend: arrows show inheritance or main object relationships. See docs/class_diagram.mmd for Mermaid source.

OOP Concepts Implemented

- Classes and Objects: All modules use C++ classes and objects.
- Encapsulation: Private data members are accessed through public methods.
- Constructors: Default and parameterized constructors are used in many classes.
- Destructors: Library and Invoice clean dynamic memory.
- Single Inheritance: Student inherits from Person.
- Multi-level Inheritance: GradStudent inherits from Student.
- Multiple Inheritance: HostelManager inherits from Accommodation and Reportable.
- Virtual Inheritance: Accommodation and Reportable virtually inherit CampusService.
- Abstract Classes: Person, LibraryItem, Accommodation, and Reportable.
- Runtime Polymorphism: Person pointers call different displayInfo functions.
- Operator Overloading: Course ==, Course <<, Course +, FeeRecord -=.
- Friend Functions: operator<< is used for Course and Invoice.
- Static Members: Invoice uses invoiceCounter.
- Copy Constructor and Assignment: FeeRecord and Invoice examples.
- Search Functions: Library has searchByTitle and searchByID.
- Array Collections: Arrays are used for courses, library items, rooms, and students.
- Exception Handling: CapacityExceededException and OverdueException.
- File I/O: Library catalog and campus report files.
- Sorting: Reports sort students by GPA.
- Composition: HostelManager contains HostelBlock.
- Aggregation: Course stores Faculty pointer and Room stores Student pointers.

Library and Finance Details

The Library module loads records from data/library_catalog.txt, displays books and journals, searches by title or ID, issues a book, shows issued records, and checks overdue returns. The Finance module tracks fee balance, accepts payment with operator-=, deep-copies payment history, and creates invoices with a static invoice counter.

Testing Summary

- Build command .\build.bat completed successfully.
- Menu options 1 to 7 were tested.
- Library demo loaded catalog data, issued B001, and showed overdue fine.
- Finance demo showed payment, copy constructor, copy assignment, invoice, and invoice copy.
- Hostel demo showed service name, allocation, duplicate check, summary, report, and vacate room.
- Reports demo sorted students by GPA and created data/campus_report.txt.
- Wrong input test with abc did not freeze the program.

Challenges and Solutions

- Understanding polymorphism: solved by using Person* array in main.cpp.
- Managing dynamic memory: solved by deleting Library items and Invoice item arrays.
- Handling full course capacity: solved by using a custom exception class.

- File handling format: solved by using simple pipe-separated text data.

Conclusion

This project helped practice core C++ OOP concepts in a practical campus system. The project is intentionally console-based and simple so each module can be explained in viva. Before final submission, real output screenshots should be added by the group.

References

- Course project brief provided by HITEC University Taxila.
- C++ class notes and lab practice.
- cppreference.com for basic C++ syntax reference.